

Towards Lag Consensus with Noisy Digital Twins Perception in Second-Order Multi-Agent Cyber-Physical Systems

Zhicheng Zhang

(Joint work with Fausto Francesco Lizzio, Zhongjun Ma, Masaaki Nagahara)

Department of Electrical Engineering
Kyoto University

The 23rd IFAC World Congress (IFAC WC)
Busan, Korea, August 25, 2026

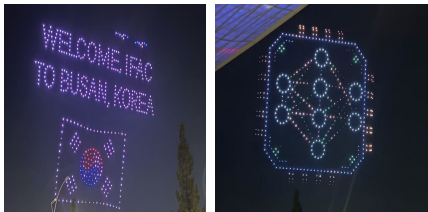
Table of Contents

- Background and Review Lag Consensus of MAS
- Problem Setup for PT-DT network modeling
- Controller Design and Stability Guarantees
- Numerical Simulations and Conclusion

What we intend to do (Goal):

Develop a framework for achieving lag consensus under noisy DT perceptions in second-order stochastic MAS

Background: Cyber-Physical Networked Systems

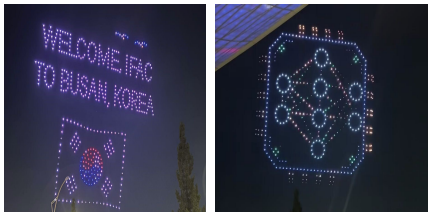


UAV formation: drone light shows



CPS: physical twins - digital twins

Background: Cyber-Physical Networked Systems



UAV formation: drone light shows



CPS: physical twins - digital twins

- MAS/CPS networked system $\implies$ ODE + Graph Theory $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$
- Physical twins (PT) - Digital twins (DT) perception/interactions $\implies$ realistic networked modeling (SDE, multi-layers, high-order, ...)
- Coordination/collective behavior $\implies$ Consensus, Synchronization
- Realization by a reliable/robust/resilient “controller design.”

Leader-following Second-order Agent Dynamics

Dynamics of second-order agents (N Followers):

$$\begin{cases} \dot{x}_i(t) = v_i(t), \\ \dot{v}_i(t) = f(x_i(t), v_i(t)) + c_1 \sum_{j \in \mathcal{N}_i} a_{ij} (x_j(t) - x_i(t)) \\ \quad + c_2 \sum_{j \in \mathcal{N}_i} a_{ij} (v_j(t) - v_i(t)) + u_i(t), \end{cases} \quad (1)$$

- $x_i(t), v_i(t), u_i(t) \in \mathbb{R}^n$ indicate the position, velocity, and control input of (follower) agent i , $\forall i \in \mathcal{V}$, respectively.

Assum. 1 (Lips. Cond.) The function $f : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is nonlinear and Lipschitz continuous in (x, v) ; i.e., there exist constants $\rho_1, \rho_2 \geq 0$

$$\|f(x_1, v_1) - f(x_2, v_2)\| \leq \rho_1 \|x_1 - x_2\| + \rho_2 \|v_1 - v_2\|, \quad \forall x, v \in \mathbb{R}^n$$

$\implies$ (Virtual) Reference/ **Leader Dynamics**:

$$\dot{x}_0(t) = v_0(t), \quad \dot{v}_0(t) = f(x_0(t), v_0(t)), \quad x_0, v_0 \in \mathbb{R}^n. \quad (2)$$

What is Lag Consensus and Why Introduce “Lag” ?

Definition (Second-order Lag Consensus, cf. Wang & Ma, 2016)

The second-order lag consensus implies the followers' states track those of the leader with a time delay $\tau \geq 0$ such that

$$\lim_{t \rightarrow +\infty} \|x_i(t) - x_0(t - \tau)\| = 0, \quad \lim_{t \rightarrow +\infty} \|v_i(t) - v_0(t - \tau)\| = 0, \quad \forall i \in \mathcal{V}.$$

What is Lag Consensus and Why Introduce “Lag” ?

Definition (Second-order Lag Consensus, cf. Wang & Ma, 2016)

The second-order lag consensus implies the followers' states track those of the leader with a time delay $\tau \geq 0$ such that

$$\lim_{t \rightarrow +\infty} \|x_i(t) - x_0(t - \tau)\| = 0, \quad \lim_{t \rightarrow +\infty} \|v_i(t) - v_0(t - \tau)\| = 0, \quad \forall i \in \mathcal{V}.$$

Remarks:

- **Motivation:** In many MAS, followers cannot respond instantaneously to the leader. Lag consensus captures delayed coordination while preserving the leader's dynamics (vs. *LF consensus* when $\tau = 0$!)
- **Merits:** By embedding the “lag” to leader, it improves the “safety” (e.g., collision avoidance & congestion prevention in tracking problem)
- **Potential App.:** Vehicle platooning tracking, UAV formation control

Problem Setup I: CPS PT-DT Networks Modeling

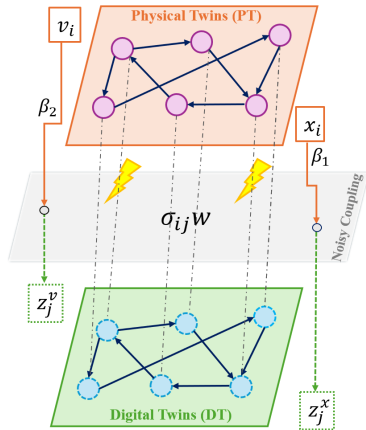
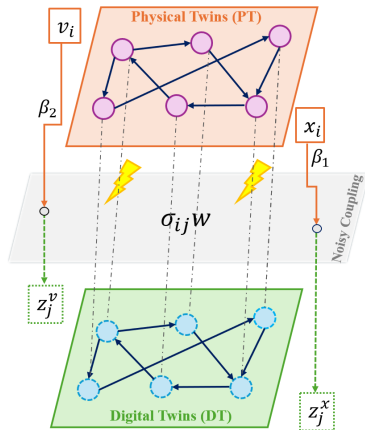


Figure: PT-DT interaction in MAS based CPS under stochastic noise (e.g., sensing jitter, disturbances, estimation errors, ...)

Problem Setup I: CPS PT-DT Networks Modeling



◇ True states x_j and v_j are perturbed as:

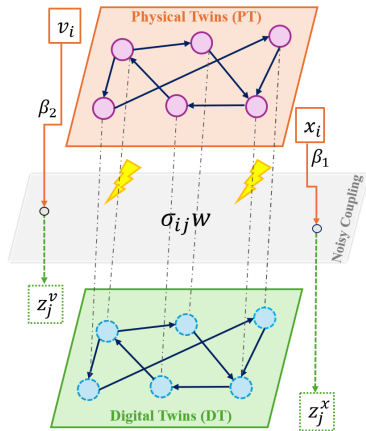
$$z_j^x(t) = x_j(t) + \beta_1 \sigma_{ij} (x_j(t) - x_i(t)) w(t),$$

$$z_j^v(t) = v_j(t) + \beta_2 \sigma_{ij} (v_j(t) - v_i(t)) w(t),$$

• $z_j^x(t), z_j^v(t)$ denotes the DT-perceived estimates of the physical states x_j, v_j .

Figure: PT-DT interaction in MAS based CPS under stochastic noise (e.g., sensing jitter, disturbances, estimation errors, ...)

Problem Setup I: CPS PT-DT Networks Modeling



◇ True states x_j and v_j are perturbed as:

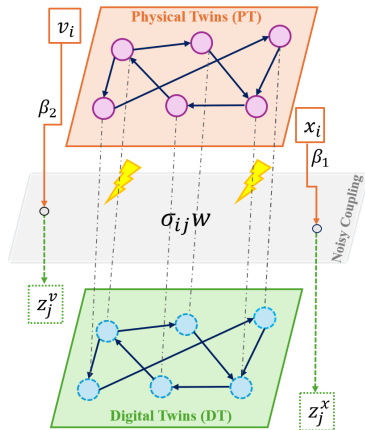
$$z_j^x(t) = x_j(t) + \beta_1 \sigma_{ij} (x_j(t) - x_i(t)) w(t),$$

$$z_j^v(t) = v_j(t) + \beta_2 \sigma_{ij} (v_j(t) - v_i(t)) w(t),$$

- $z_j^x(t), z_j^v(t)$ denotes the DT-perceived estimates of the physical states x_j, v_j .
- $\beta_1, \beta_2 \in \mathbb{R}$ are the noise intensities.

Figure: PT-DT interaction in MAS based CPS under stochastic noise (e.g., sensing jitter, disturbances, estimation errors, ...)

Problem Setup I: CPS PT-DT Networks Modeling



◇ True states x_j and v_j are perturbed as:

$$z_j^x(t) = x_j(t) + \beta_1 \sigma_{ij} (x_j(t) - x_i(t)) w(t),$$

$$z_j^v(t) = v_j(t) + \beta_2 \sigma_{ij} (v_j(t) - v_i(t)) w(t),$$

- $z_j^x(t), z_j^v(t)$ denotes the DT-perceived estimates of the physical states x_j, v_j .
- $\beta_1, \beta_2 \in \mathbb{R}$ are the noise intensities.
- A white noise $w(t) \in \mathbb{R}$ satisfies $\int_0^t w(s) ds = W(t)$, $W(t)$ is a standard Wiener process. The gains $\sigma_{ij} > 0$ iff $a_{ij} > 0$, and $\sigma_{ij} = 0$ otherwise.

Figure: PT-DT interaction in MAS based CPS under stochastic noise (e.g., sensing jitter, disturbances, estimation errors, ...)

Problem Setup I: CPS PT-DT Networks Modeling

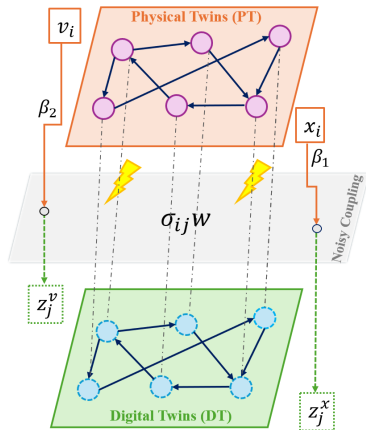


Figure: PT-DT interaction in MAS based CPS under stochastic noise (e.g., sensing jitter, disturbances, estimation errors, ...)

◇ True states x_j and v_j are perturbed as:

$$z_j^x(t) = x_j(t) + \beta_1 \sigma_{ij} (x_j(t) - x_i(t)) w(t),$$

$$z_j^v(t) = v_j(t) + \beta_2 \sigma_{ij} (v_j(t) - v_i(t)) w(t),$$

- $z_j^x(t), z_j^v(t)$ denotes the DT-perceived estimates of the physical states x_j, v_j .
- $\beta_1, \beta_2 \in \mathbb{R}$ are the noise intensities.
- A white noise $w(t) \in \mathbb{R}$ satisfies $\int_0^t w(s) ds = W(t)$, $W(t)$ is a standard Wiener process. The gains $\sigma_{ij} > 0$ iff $a_{ij} > 0$, and $\sigma_{ij} = 0$ otherwise.

♣ Noise enters through relative DT-PT:

$$c_1 \sum_{j \in \mathcal{N}_i} a_{ij} (z_j^x(t) - x_i(t)), c_2 \sum_{j \in \mathcal{N}_i} a_{ij} (z_j^v(t) - v_i(t)).$$

Problem Setup II: Mean-square Lag Consensus

Definition (Mean-square lag consensus)

For a predefined lag $\tau \geq 0$, the LF multi-agent cyber-physical network under PT-DT interaction is said to achieve second-order mean-square lag consensus if, for all $i \in \mathcal{V}$ and any initial states

$$\lim_{t \rightarrow \infty} \mathbb{E} \|\tilde{x}_i(t)\| = 0, \quad \lim_{t \rightarrow \infty} \mathbb{E} \|\tilde{v}_i(t)\| = 0, \quad \implies \quad \xi_i = (\tilde{x}_i^\top, \tilde{v}_i^\top)^\top.$$

- Lag errors $\tilde{x}_i(t) = x_i(t) - x_0(t_\tau)$ and $\tilde{v}_i(t) = v_i(t) - v_0(t_\tau)$ with $t_\tau = t - \tau$.

Problem

Under specific Assumptions and stochastic PT-DT interactions, design a resilient protocol $u_i(t)$ to achieve mean-square second-order lag consensus.

Key Issue:

- How to appropriately characterize/design the resilient control inputs? (e.g., lag errors + stochastic coupling by noisy PT-DT perception)

Lag Consensus Protocol Design

We design a piecewise, resilient intermittent (on/off) control policy:

$$u_i(t) = \begin{cases} -\kappa_i \left(c_1 (x_i(t) - x_0(t - \tau)) + c_2 (v_i(t) - v_0(t - \tau)) \right) \\ - \sum_{j \in \mathcal{N}_i} \sigma_{ij} \left(\beta_1 (x_i(t) - x_j(t)) + \beta_2 (v_i(t) - v_j(t)) \right) w(t), & t \in [t_k, s_k], \\ 0, & t \in (s_k, t_{k+1}), \end{cases}$$

Assum. 2 (Intermittent Input Failures) For the intermittent input failures (possibly not periodic), there exist two scalars θ and δ such that

$$\inf_{k \in \mathbb{N}} (s_k - t_k) = \theta, \quad \sup_{k \in \mathbb{N}} (t_{k+1} - t_k) = \delta, \quad 0 < \theta < \delta < +\infty.$$

★ Define the maximal ratio of time with control input failures

$$\psi = \lim_{k \rightarrow +\infty} \sup_{k \in \mathbb{N}} \frac{t_{k+1} - s_k}{t_{k+1} - t_k}, \quad \implies \quad \psi \leq 1 - \frac{\theta}{\delta}, \quad \psi \in (0, 1).$$

- $[t_k, s_k]$ is the *control-active* period; (s_k, t_{k+1}) is the *inactive* period.

Recast Stochastic Lag Consensus as Stability of SDE

Proposition (Recasting Lag Error Dynamics via SDE) The stochastic error dynamics can be recast as the stochastic differential equation (SDE):

$$\begin{cases} d\xi(t) = (\tilde{F}_\tau(t) + (H_1 \otimes I_n)\xi(t))dt - (U \otimes I_n)\xi(t)dW(t), & t \in [t_k, s_k], \\ d\xi(t) = (\tilde{F}_\tau(t) + (H_2 \otimes I_n)\xi(t))dt, & t \in (s_k, t_{k+1}). \end{cases}$$

where $\tilde{L} = L + K$ with $K = \text{diag}\{\kappa_1, \dots, \kappa_\ell, 0, \dots, 0\}$ for $\kappa_i \geq 0$, and $H_1 = \begin{pmatrix} 0_N & I_N \\ -c_1 \tilde{L} & -c_2 \tilde{L} \end{pmatrix}$, $H_2 = \begin{pmatrix} 0_N & I_N \\ -c_1 L & -c_2 L \end{pmatrix}$, $\tilde{F}_\tau(t) = \begin{pmatrix} 0_{nN} \\ F(x(t), v(t)) - \mathbf{f}_N(x_0^\tau, v_0^\tau) \end{pmatrix}$, $F(x, v) = (f(x_1, v_1)^\top, \dots, f(x_N, v_N)^\top)^\top$, $\mathbf{f}_N(x_0^\tau, v_0^\tau) = \mathbf{1}_N \otimes f((x_0(t_\tau), v_0(t_\tau)))$.

Facts:

- Reaching lag consensus Problem equals ensuring the mean-square stability of stochastic error dynamics.
- Only a few agents require control (a.k.a., pinning control, $1 \leq \ell \leq N$)
see cf. Øksendal, 2003; Porfiri & Bernardo, 2008; Liu & Chen, 2016

Theorem

Based on stochastic error dynamics, the control protocol $u_i(t)$ solves lag consensus Problem, that is, ensures mean-square stability and achieves second-order lag consensus of CPS, if the next conditions are satisfied:

- (i) $\gamma := 2\lambda_{\min}(R_1) - \lambda_{\max}(U^\top P U) \geq 0$,
- (ii) $\bar{\mu}_1\theta - \bar{\mu}_2(\delta - \theta) > 0$,

where $\bar{\mu}_1 = \gamma/\lambda_{\max}(P_2)$, $\bar{\mu}_2 = 2\lambda_{\max}(R_2)/\lambda_{\min}(P_1)$, and the parameters and matrices P , P_1 , P_2 , R_1 , and R_2 are well-defined.

Theorem

Based on stochastic error dynamics, the control protocol $u_i(t)$ solves lag consensus Problem, that is, ensures mean-square stability and achieves second-order lag consensus of CPS, if the next conditions are satisfied:

- (i) $\gamma := 2\lambda_{\min}(R_1) - \lambda_{\max}(U^\top P U) \geq 0$,
- (ii) $\bar{\mu}_1\theta - \bar{\mu}_2(\delta - \theta) > 0$,

where $\bar{\mu}_1 = \gamma/\lambda_{\max}(P_2)$, $\bar{\mu}_2 = 2\lambda_{\max}(R_2)/\lambda_{\min}(P_1)$, and the parameters and matrices P , P_1 , P_2 , R_1 , and R_2 are well-defined.

Lyapunov Stability for SDE + Ito formula (see cf. Øksendal, 2003):

$$V(\xi) = \frac{1}{2}\xi(t)^\top (P \otimes I_n)\xi(t), \quad P = \begin{pmatrix} 2c_1c_2\tilde{L}^s & c_1I_N \\ c_1I_N & c_2I_N \end{pmatrix}$$

- An analytical, physical-informed (system level) P is designed here!

- Exponentially convergence: $\frac{1}{t} \log(\mathbb{E}\|\xi(t)\|^2) \leq \frac{1}{t} \left(\log \frac{2\bar{M}}{\lambda_{\min}(P_1)} - \check{\mu}t \right) \rightarrow -\check{\mu} < 0$

Numerical Simulations (Small World Networks)

- Small world (SW) network is simulated as
$$f(x_i, v_i) = -\sin(x_i) - 0.15v_i - 0.2, \quad i = 0, \dots, 50.$$
- Agent 0 is the leader, and only agent 1 receives delayed information from the leader. Time horizon: $t \in [0, 30]$ s; failure ratio $\psi = 0.7$.
- Random intermittent failures: $T_k \sim \mathcal{U}[0.3, 1.0]$, $T_k^c \sim \mathcal{U}[0.3, T_k]$.
- Parameters: $c_1 = c_2 = 1$, $\beta_1 = \beta_2 = 0.9$, $\rho_1 = 1$, $\rho_2 = 0.15$, $\tau = 2$.
- Vary κ_1 to investigate its effect on lag consensus.

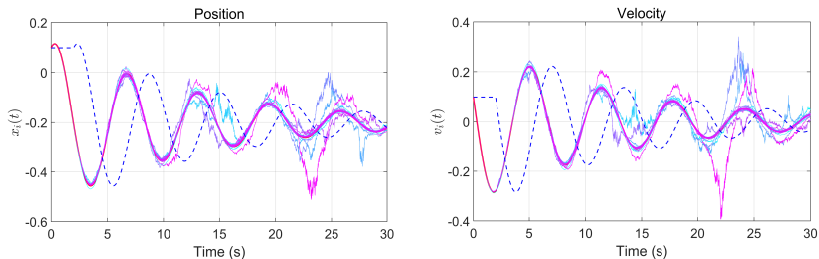


Figure: Lag consensus with small feedback gain $\kappa_1 = 1.7$ (Not Good).

Exponential Convergence

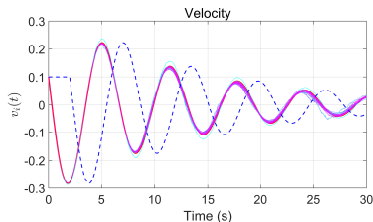
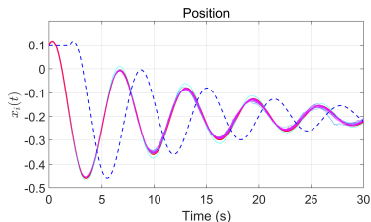


Figure: Lag consensus with feedback gain $\kappa_1 = 5.5$ (good convergence)

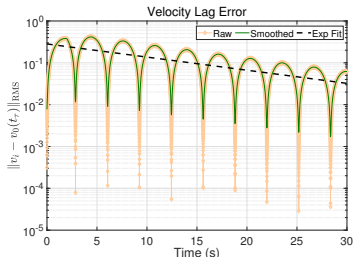
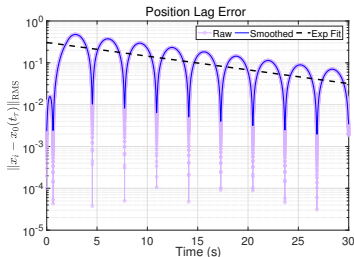
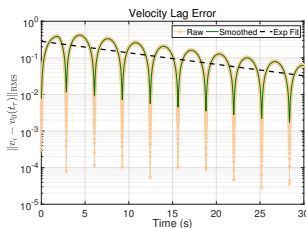
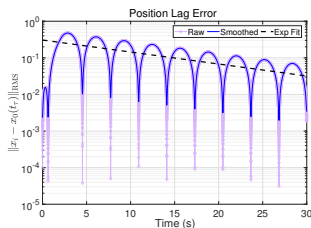


Figure: Lag errors are both decay exponentially in the mean-square sense.

The take-home message:

- **Modeling:** Second-order lag consensus with PT-DT interactions.
- **Resilience:** Stochastic noise and intermittent input failures.
- **Mean-square stability:** Stability of the resulting SDE error dynamics.
- **Simulations:** Small-world CPS network simulations.
- **Future works:** controller's performance, the effect of delay margins.



arXiv:2605.04692

